

GENERAL HOSPITAL — ECHOCARDIOGRAPHY (ECO) UNIT

Intelligent Patient Queue Management and Prediction System

PATIENT (USER) MOBILE APPLICATION

Screen-by-Screen Interface & Functional Description

Primary Actor: Patient / End-User of the ECO Unit Patient Portal

Interface: Patient Mobile Application • 14 implemented screens

Companion reference document to the system research proposal / thesis

1. Overview

The system proposed in the accompanying research is designed around three coordinated interfaces — a Patient (User) Mobile Application, a Staff Web Portal, and an Administrator Web Portal — built to manage patient flow and predict queue behaviour in the Echocardiography (ECO) unit of a Sri Lankan public hospital. This document focuses exclusively on the Patient Mobile Application, the patient-facing side of that system, and provides a detailed, screen-by-screen description of its user interface and functionality.

The Patient Mobile Application is used by a single actor — the Patient (also referred to as the End-User or Registered Patient) — across every screen described below. It is designed to replace manual, paper-based registration and queue handling with self-service digital registration, machine-learning-based wait-time prediction, live position tracking, and GPS/QR-verified on-site priority handling. Fourteen implemented screens are covered: five relate to secure account access, three relate to profile and security management, five form the core queue journey, and one manages on-site priority verification.

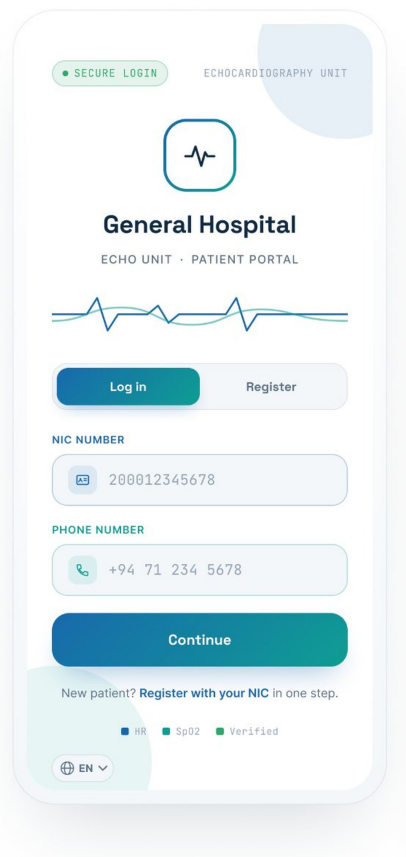
Each entry below states the screen's purpose, its key interface elements exactly as implemented in the design, how it functions within the wider system, and the research objective or literature theme it corresponds to in the study.

1.1 Screen Index

#	Screen Name	Module / Category
1	Secure Login	Authentication / Access Module
2	New Patient Registration	Authentication / Access Module
3	Verify OTP	Authentication / Access Module
4	Verification Successful	Authentication / Access Module
5	Update Phone Number	Profile & Security Module
6	Home Dashboard	Core Patient Journey
7	Join Today's Queue	Core Patient Journey
8	Live Ticket ("Your Number")	Core Patient Journey
9	Live Queue Tracking	Core Patient Journey
10	Visit History	Core Patient Journey
11	Profile	Profile & Security Module
12	Edit Profile	Profile & Security Module
13	Check-in QR	Priority & Verification Module
14	Notifications	Core Patient Journey

2. Screen-by-Screen Descriptions

Screen 1 | Secure Login



Actor: Registered Patient (returning user)

Module / Category: Authentication / Access Module

Purpose

The entry-point screen of the Patient Mobile Application. It allows an already-registered ECO-unit patient to securely sign in to the portal using their National Identity Card (NIC) number and registered mobile number, instead of a conventional username/password pair.

Key Interface Elements

- Status badge “SECURE LOGIN” and unit label “ECHOCARDIOGRAPHY UNIT”
- Hospital brand mark (heart-rate/pulse icon), “General Hospital” title and “ECHO Unit · Patient Portal” subtitle
- Animated ECG waveform graphic reinforcing the cardiology/ECO context
- “Log in / Register” segmented tab control (Log in selected)
- NIC Number input field with ID-card icon
- Phone Number input field, pre-formatted for Sri Lankan mobile numbers (+94)
- Gradient “Continue” call-to-action button
- “New patient? Register with your NIC in one step” shortcut link
- Footer legend (HR / SpO2 / Verified) and a language selector (EN)

Functionality & System Behaviour

On submission, the entered NIC and phone number are validated against the patient record; a successful match triggers an SMS one-time-password sent to the registered number and routes the patient to the Verify OTP screen. This NIC-based login removes the need to remember a separate password, reduces duplicate

patient records, and directly supports the study's objective of replacing manual, paper-based registration at the ECO unit with a secure digital entry point.

Research Link

Supports Research Objective 2 (digital registration module) and addresses research gap G1 identified in Chapter 2 of the study.

Screen 2 | New Patient Registration

The image shows a mobile application interface for a new patient registration screen. At the top, there is a status bar with a green pill-shaped badge containing a white heart icon and the text "NEW PATIENT", and a unit label "ECHOCARDIOGRAPHY UNIT". Below this is a circular icon with a white heart and a black ECG line. The main heading is "General Hospital" in bold, followed by "ECHO UNIT · PATIENT PORTAL" in a smaller font. A decorative ECG line graphic is positioned below the heading. There are two buttons: "Log in" (white with a grey border) and "Register" (solid green with white text). Below the buttons are four input fields, each with a label and a corresponding icon: "FULL NAME" (magnifying glass icon) with the text "Kasun Perera", "NIC NUMBER" (ID card icon) with the text "200012345678", "PHONE NUMBER" (phone icon) with the text "+94 71 234 5678", and "DATE OF BIRTH" (calendar icon) with the text "DD / MM / YYYY". At the bottom, there is a large green button with the text "Create account →". Below this button, there is a link: "Already registered? Log in with your NIC instead." At the very bottom, there is a status bar with three indicators: "HR" (blue square), "SpO2" (green square), and "Verified" (green square).

Actor: First-time / walk-in Patient

Module / Category: Authentication / Access Module

Purpose

A self-service sign-up screen that lets a first-time patient create their ECO-unit portal account directly from their phone, without needing a staff member to manually enter their details at the counter.

Key Interface Elements

- Status badge "NEW PATIENT" and unit label
- "Log in / Register" tab control (Register selected)
- Full Name input field
- NIC Number input field
- Phone Number input field (+94 format)
- Date of Birth input field (DD / MM / YYYY)

- Gradient “Create account →” button
- “Already registered? Log in with your NIC instead” link

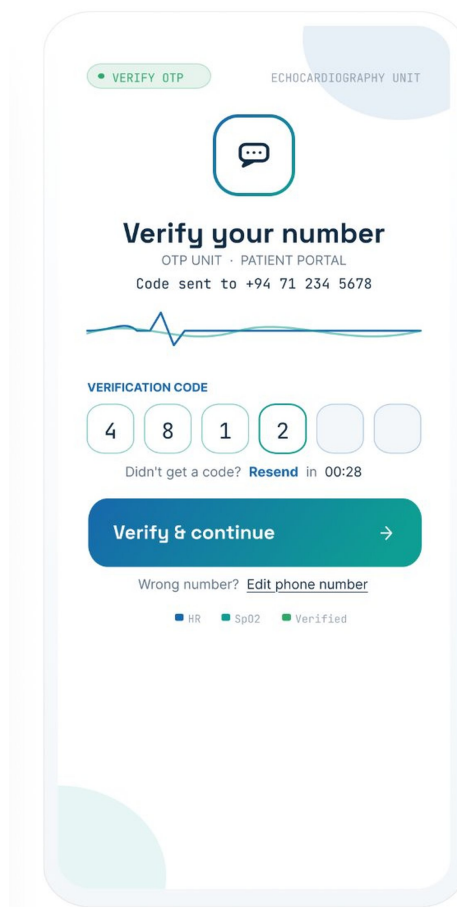
Functionality & System Behaviour

Captures the minimum demographic data set (name, NIC, phone, date of birth) needed to create a unique patient record tied to the NIC number, preventing duplicate or misplaced records that are common under manual registration. Submitting the form triggers the same OTP verification flow (Screen 3) as login, confirming the patient controls the phone number before the account is activated.

Research Link

Directly operationalises Research Objective 2 — the digital patient-registration module — and reduces the administrative overhead identified in the Chapter 1 problem statement.

Screen 3 | Verify OTP



Actor: Patient (mid login or registration)

Module / Category: Authentication / Access Module

Purpose

A two-factor security checkpoint that confirms the patient genuinely controls the mobile number they logged in or registered with, before the app grants access to any queue or personal data.

Key Interface Elements

- Status badge “VERIFY OTP”, message-bubble icon
- “Verify your number” heading and “Code sent to +94 71 234 5678” confirmation text
- Six-digit segmented OTP input boxes
- “Didn’t get a code? Resend in 00:28” countdown / resend control
- Gradient “Verify & continue →” button

- “Wrong number? Edit phone number” escape link

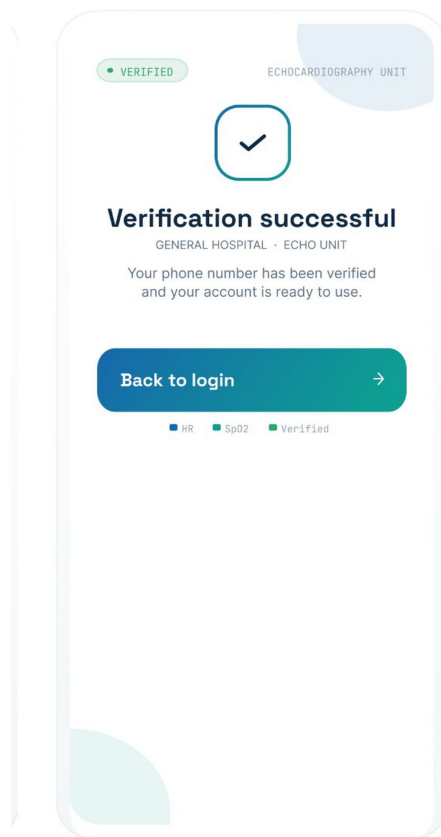
Functionality & System Behaviour

Validates the time-limited one-time code sent by SMS. This protects patient accounts from unauthorised access and, because the verified number becomes the channel for all later alerts, guarantees that queue-position and check-in notifications (see Screen 14) reliably reach the correct patient.

Research Link

Security control underpinning Objectives 2 and 5; mirrors the SMS-based verification pattern reported in the local OPD-PMMAS and Doc990 literature reviewed in Chapter 2.

Screen 4 | Verification Successful



Actor: Patient

Module / Category: Authentication / Access Module

Purpose

A short confirmation screen that closes the login/registration/OTP flow, reassuring the patient that their phone number has been verified and their account is ready to use.

Key Interface Elements

- Status badge “VERIFIED” and checkmark icon
- “Verification successful” heading with “General Hospital · Echo Unit” subtitle
- Confirmation message: “Your phone number has been verified and your account is ready to use.”
- Gradient “Back to login →” button

Functionality & System Behaviour

Provides positive, unambiguous closure to the authentication sequence rather than silently redirecting the patient, which reduces confusion for first-time or less digitally confident users. The button returns the patient

to the Secure Login screen (or forward into the Home Dashboard, depending on the entry flow) so they can proceed to use the queue features.

Research Link

Usability safeguard supporting the digital-literacy considerations discussed in Chapter 5 (Section 5.9) of the study.

Screen 5 | Update Phone Number

The screen is titled "Update phone number" under the heading "PATIENT PORTAL · SECURITY". It features a status bar at the top with "EDIT DETAILS" and "ECHOCARDIOGRAPHY UNIT". A phone icon is centered above the title. Below the title, explanatory text states: "We'll text a one-time code to confirm any changes to this number." The form includes two input fields: "CURRENT NUMBER" (displaying "+94 71 234 5678") and "NEW PHONE NUMBER" (with a placeholder "Enter new number"). A prominent teal button labeled "Send verification code" with a right arrow is positioned below the new number field. A link "Cancel and go back" is located beneath the button. At the bottom, a progress indicator shows three items: "HR" (blue square), "SpO2" (teal square), and "Verified" (green square).

Actor: Patient

Module / Category: Profile & Security Module

Purpose

A dedicated security screen that lets a patient change the mobile number linked to their account — for example after switching SIM cards — since that number is the primary channel for OTP login and live queue alerts.

Key Interface Elements

- Status badge "EDIT DETAILS", phone icon, "Update phone number" heading, "Patient Portal · Security" subtitle
- Explanatory text: "We'll text a one-time code to confirm any changes to this number."
- Read-only "Current number" field
- Editable "New phone number" field
- Gradient "Send verification code →" button
- "Cancel and go back" link

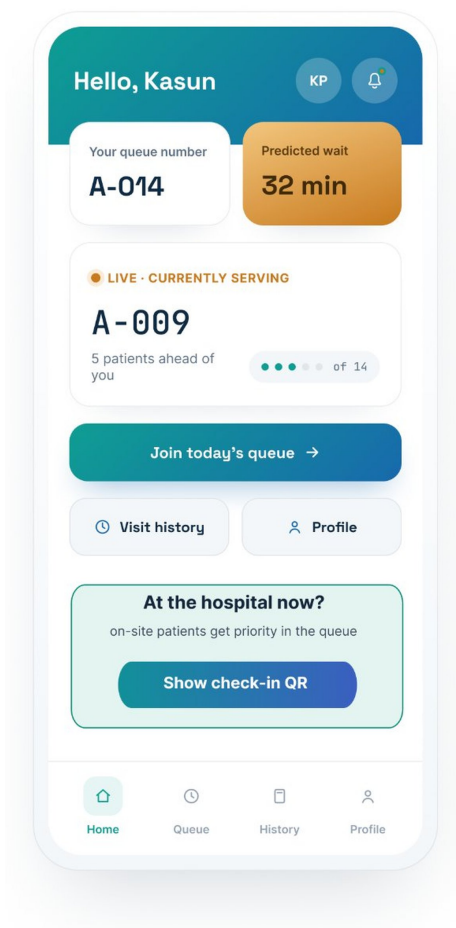
Functionality & System Behaviour

Any change to the registered number must be re-confirmed by OTP before it takes effect, preventing an unauthorised party from silently hijacking the account and intercepting queue notifications. This screen is reached from the Profile screen's "Edit details" action.

Research Link

Protects the reliability of the real-time notification channel used by Screens 6, 9 and 14.

Screen 6 | Home Dashboard



Actor: Patient (logged in)

Module / Category: Core Patient Journey

Purpose

The patient's central landing screen after login. It summarises the patient's current ticket, the machine-learning-predicted wait time, and provides one-tap access to every other core action, so the patient never has to ask staff "how much longer?"

Key Interface Elements

- Personalised gradient header "Hello, Kasun" with avatar initials (KP) and a notification bell showing an unread indicator
- "Your queue number" card (e.g. A-014)
- ML-generated "Predicted wait" badge (e.g. 32 min) in an amber gradient card
- "LIVE · Currently serving" card showing the ticket currently being attended (A-009), "5 patients ahead of you" and a dot-progress indicator ("of 14")
- Primary gradient "Join today's queue →" button
- Secondary "Visit history" and "Profile" quick-access buttons

- “At the hospital now?” priority callout explaining that on-site patients get priority in the queue, with a “Show check-in QR” button
- Bottom navigation bar: Home / Queue / History / Profile

Functionality & System Behaviour

Surfaces the output of the system’s wait-time prediction model (Random Forest / XGBoost, Section 3.8 of the thesis) and the live queue position as an always-available, self-service data point on the patient’s own device, replacing tacit, staff-mediated updates. It is also the launch point for joining the queue and for the on-site QR check-in that unlocks priority handling.

Research Link

Central implementation of Research Objectives 2, 4 and 5 (registration, ML wait-time prediction and real-time status communication).

Screen 7 | Join Today’s Queue

Actor: Patient

Module / Category: Core Patient Journey

Purpose

The digital ticket-request screen where a patient selects their reason for visiting and a preferred time slot, generating a live queue ticket without needing to visit the registration counter first — the digital replacement for the paper queue slip.

Key Interface Elements

- Back navigation and “Join today’s queue” title
- “Reason for visit” dropdown selector (e.g. “Routine ECHO scan”)

- “Preferred time slot” chip selector (9–11 AM / 11–1 PM / 2–4 PM)
- “Notes for staff (optional)” free-text field for symptoms or referral details
- ML-driven “Estimated wait for this slot: 40 minutes” callout
- Gradient “Confirm and join queue” button
- Reassurance note: “You can leave the waiting room — we’ll track your place automatically.”

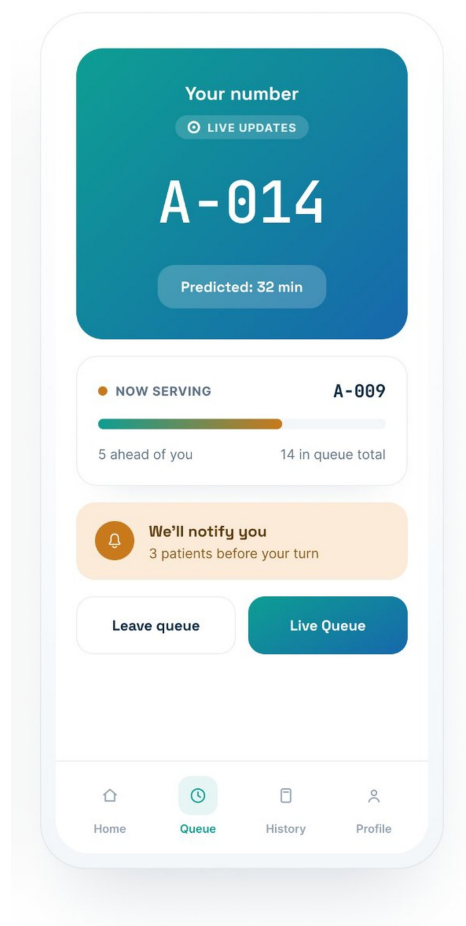
Functionality & System Behaviour

Submitting the form creates a new digital queue ticket and feeds the selected service type, time slot and notes into the queue-prediction pipeline, which returns an estimated wait time for that specific slot before the patient commits. This lets patients register and effectively “queue remotely” instead of waiting physically at the unit from the moment they arrive.

Research Link

Implements Research Objective 2 (automated queue-ticket generation) and the department/service selector referenced in the Chapter 5 discussion of the Patient Mobile Application.

Screen 8 | Live Ticket (“Your Number”)



Actor: Patient

Module / Category: Core Patient Journey

Purpose

An immediate confirmation screen shown right after a ticket is issued, giving the patient their official queue number and a live snapshot of the queue.

Key Interface Elements

- Gradient header card “Your number” with a “LIVE UPDATES” badge and the large assigned ticket number (e.g. A-014)
- “Predicted: 32 min” badge
- “NOW SERVING” card with the currently-called ticket (A-009) and a visual progress bar
- “5 ahead of you” and “14 in queue total” counters
- “We’ll notify you — 3 patients before your turn” alert
- “Leave queue” (secondary) and “Live Queue” (primary) action buttons

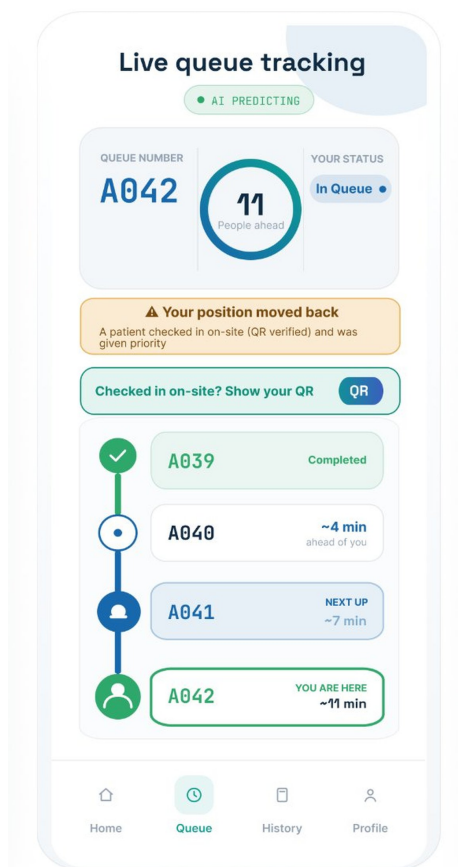
Functionality & System Behaviour

Confirms that the ticket has been successfully created and re-states the ML-predicted wait time so the patient has an immediate expectation to plan around. The “Live Queue” button takes the patient to the full, staged Live Queue Tracking screen (Screen 9) for a more detailed view.

Research Link

Bridges Research Objectives 2 and 5, connecting ticket issuance to real-time tracking.

Screen 9 | Live Queue Tracking



Actor: Patient

Module / Category: Core Patient Journey

Purpose

The most detailed real-time screen in the application: a colour-coded, stage-by-stage timeline that shows exactly where the patient sits in the queue, continuously refreshed by the prediction engine, and transparently explains why their position may have changed.

Key Interface Elements

- “Live queue tracking” title with an “AI PREDICTING” status badge

- Queue-number card (e.g. A042) with a circular “People ahead” counter (11) and a “Your status: In Queue” tag
- Amber “Your position moved back” alert explaining that another patient checked in on-site (QR verified) and was given priority
- “Checked in on-site? Show your QR” shortcut button linking to Screen 13
- Vertical staged timeline of surrounding tickets, each with a colour-coded status icon: completed (green check, e.g. A039), in progress (pulsing dot, e.g. A040, “~4 min ahead of you”), next up (staff icon, e.g. A041, “~7 min”), and the patient’s own ticket highlighted as “YOU ARE HERE” (e.g. A042, “~11 min”)
- Bottom navigation bar

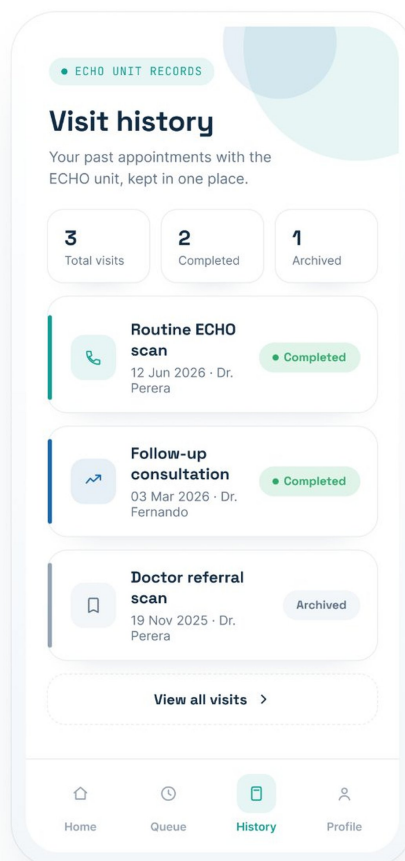
Functionality & System Behaviour

This screen is the direct interface implementation of the system’s priority-based queue-management logic: it does not just show a number, it visually explains that a legitimately prioritised on-site check-in can move a remote patient’s position back, so the patient understands the change instead of assuming an error. Position and wait estimates update live from the ML prediction service as tickets are called, served, or reprioritised by staff on the Queue Management screen of the Staff Portal.

Research Link

Core implementation of Research Objectives 3, 4 and 5 (priority-based queue management, ML wait-time/volume prediction, real-time tracking) and closes research gap G2 discussed in Chapter 5.

Screen 10 | Visit History



Actor: Patient

Module / Category: Core Patient Journey

Purpose

Gives the patient self-service access to a record of their past attendances at the ECO unit, kept in one place instead of on paper slips or in staff-side files only.

Key Interface Elements

- “ECHO UNIT RECORDS” badge, “Visit history” title and subtitle
- Summary stat cards: “3 Total visits”, “2 Completed”, “1 Archived”
- A list of visit cards, each with a coloured status stripe, an icon, the visit type (e.g. Routine ECHO scan, Follow-up consultation, Doctor referral scan), the date and attending doctor, and a status pill (Completed / Archived)
- “View all visits” link
- Bottom navigation bar

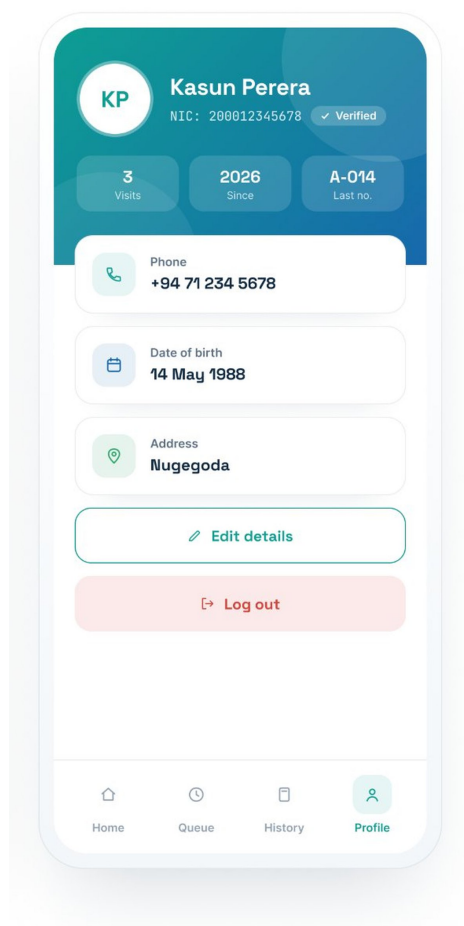
Functionality & System Behaviour

Lets the patient review when they last attended the unit, which doctor they saw, and the outcome status of each visit, supporting continuity of care and giving the patient a transparent, self-service alternative to asking reception for their attendance record.

Research Link

One of the five core patient screens defined in the study's scope (Section 1.5).

Screen 11 | Profile



Actor: Patient

Module / Category: Profile & Security Module

Purpose

A read-only account-summary screen showing the patient's key personal and account details at a glance, and acting as the gateway to profile editing and logout.

Key Interface Elements

- Gradient header card with avatar initials (KP), full name, NIC number and a “Verified” badge
- Quick-stat chips: Visits (3), Since (2026), Last ticket number (A-014)
- Detail rows for Phone, Date of birth and Address, each with an icon
- “Edit details” button
- “Log out” button (styled in a warning/red tone)
- Bottom navigation bar

Functionality & System Behaviour

Serves as the account hub: “Edit details” opens the Edit Profile screen (Screen 12) for updating personal information, while any phone-number change is routed through the dedicated Update Phone Number security flow (Screen 5) rather than being editable directly here, since the number also serves as the OTP and notification channel.

Research Link

Supports account transparency and trust, reinforcing the “Verified” identity status established during registration/login.

Screen 12 | Edit Profile

The screenshot displays the 'Edit Profile' screen within a mobile application. At the top, a blue gradient header card features a circular avatar with the initials 'KP' and a small camera icon. Below the header, the screen is organized into several form sections, each with a title and a corresponding icon: 'Full name' (person icon) with the value 'Kasun Perera'; 'NIC number' (ID card icon) with the value '200012345678'; 'Phone' (phone handset icon) with the value '+94 71 234 5678'; 'Date of birth' (calendar icon) with the value '14 May 1988'; and 'Address' (location pin icon) with the value 'Nugegoda'. Each field has a small edit icon (pencil) to its right. At the bottom of the form area are two buttons: a blue 'Save changes' button and a white 'Cancel' button. The very bottom of the screen shows a navigation bar with four icons: a house for 'Home', a clock for 'Queue', a document for 'History', and a person for 'Profile' (which is currently selected and highlighted in green).

Actor: Patient

Purpose

An editable version of the Profile screen that lets the patient keep their personal details and profile photo up to date.

Key Interface Elements

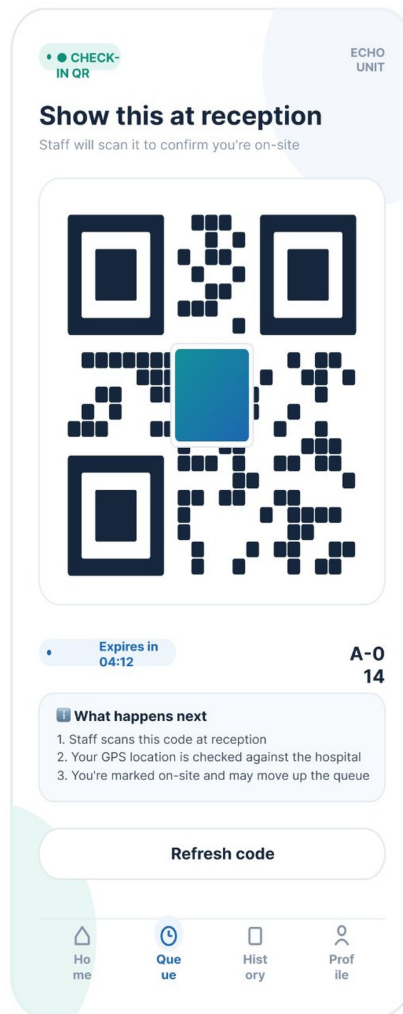
- Gradient header with the patient's avatar and a camera/upload icon for changing the profile photo
- Editable "Full name" field
- Locked/read-only "NIC number" field (padlock icon) — cannot be changed by the patient
- Editable "Phone" field
- Editable "Date of birth" field
- Editable "Address" field
- Gradient "Save changes" button and a "Cancel" button
- Bottom navigation bar

Functionality & System Behaviour

Allows the patient to correct or update contact and demographic information so SMS alerts and identity matching stay accurate, while deliberately locking the NIC number — the unique record key — to preserve the integrity of the patient database and prevent accidental duplication of records.

Research Link

Reinforces the data-integrity safeguards discussed alongside the digital-registration objective (Objective 2).



Actor: Patient (on arrival at the hospital)

Module / Category: Priority & Verification Module

Purpose

An on-site presence-verification screen shown at the hospital reception or ECO waiting area, used to prove the patient has physically arrived and to unlock the queue-priority benefit reserved for on-site patients.

Key Interface Elements

- “CHECK-IN QR” badge and “Show this at reception” heading
- Instruction text: “Staff will scan it to confirm you’re on-site”
- Large scannable QR code unique to the patient’s ticket
- “Expires in 04:12” countdown badge and the associated ticket number (A-014)
- “What happens next” info panel: (1) Staff scans this code at reception, (2) Your GPS location is checked against the hospital, (3) You’re marked on-site and may move up the queue
- “Refresh code” button
- Bottom navigation bar

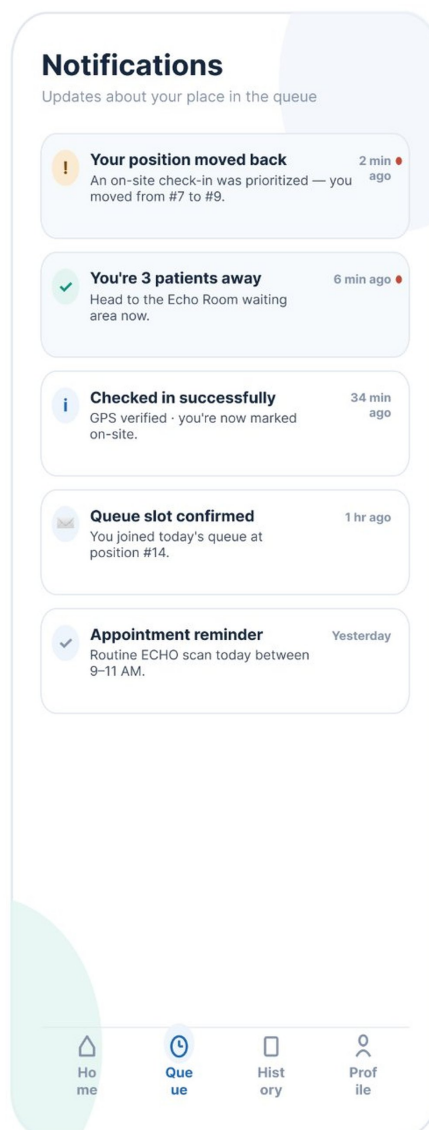
Functionality & System Behaviour

Combines a time-limited QR code with GPS geofencing so the system can reliably distinguish patients who are physically present at the ECO unit from those queueing remotely. A verified on-site check-in feeds directly into the priority-queue logic visualised on the Live Queue Tracking screen (Screen 9) and referenced on the Home Dashboard (Screen 6), giving staff and the system a trustworthy trigger for reordering the queue.

Research Link

Directly implements Research Objective 3 (priority-based queue management) and reduces queue-position disputes referenced in the Chapter 1 problem statement.

Screen 14 | Notifications



Actor: Patient

Module / Category: Core Patient Journey

Purpose

A chronological feed of every system alert relevant to the patient's current and recent queue journey, so the patient stays informed without needing to keep the Live Queue Tracking screen open at all times.

Key Interface Elements

- "Notifications" title and subtitle "Updates about your place in the queue"
- "Your position moved back" alert (priority-reorder notice, e.g. 2 min ago)
- "You're 3 patients away" proximity alert (e.g. 6 min ago)
- "Checked in successfully" GPS-verified confirmation (e.g. 34 min ago)
- "Queue slot confirmed" ticket-joined confirmation (e.g. 1 hr ago)
- "Appointment reminder" (e.g. Yesterday)
- Bottom navigation bar

Functionality & System Behaviour

Implements the push-style, always-available communication channel that the study identifies as central to reducing patients' need to repeatedly ask staff about their turn. Unread items are flagged with a red dot and mirrored by the notification bell badge on the Home Dashboard, ensuring priority changes and check-in confirmations are never missed.

Research Link

Directly supports Research Objective 5 (real-time queue tracking and status communication) and mirrors the SMS-alert pattern documented in the OPD-PMMAS prototype reviewed in Chapter 2.